

# Katalin Berlow

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## Education

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### UC Berkeley

*Ph.D. in Mathematics*

Focus on descriptive combinatorics and distributed algorithms. See [here](#) for papers.

Berkeley, CA

*Aug. 2021 – Aug. 2026*

### Carnegie Mellon University

*Bachelor of Science in Mathematics with Honors*

Pittsburgh, PA

*Aug. 2017 – May 2021*

## Experience

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### Member of Technical Staff

*0xPARC Cryptography*

Aug. 2026 – Present

*San Francisco, CA*

- Developing protocols to verify accelerator resource use during matrix multiplication under adversarial implementations.
- Investigating protocols in which reducing stored workload data forces detectable additional computation or data transfer.

### Cryptography Intern

*0xPARC Cryptography*

Jun. 2026 – Aug. 2026

*San Francisco, CA*

- Designed and implemented CKKS homomorphic algorithms for private fuzzy matching against databases of DNA strings; used ciphertext rotations and low-degree polynomials for insertion/deletion tolerance.
- Prototyped privacy-preserving nearest neighbor retrieval by computing cosine similarity scores over CKKS-encrypted vectors.

### Mathematical Problem Contributor

*Frontier Math for Epoch AI*

Apr. 2025 – May 2025

*Berkeley, CA*

- Authored a graduate-level combinatorics task for FrontierMath to probe multi-step mathematical reasoning; refined the statement and solution to eliminate ambiguity and unintended shortcut solutions.

## Technical Skills

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**Programming:** Python, NumPy.

**Tools:** Git/GitHub, Linux/Unix, command line.

**Mathematics:** Combinatorics, graph theory, algorithms and complexity, probability, measure theory, dynamical systems, ergodic theory.

## Honors and Research Programs

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**Simons Dissertation Fellowship in Mathematics**

2025

**NSF Graduate Research Fellowship**

2021

**Fulbright** (not undertaken)

2021

**Benter Foundation Mathematics Scholarship**

2020

**Duluth REU**

2020